Adaptive State–Reasoning Co-Design under Matched Total Compute

ORION-P12

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Abstract

Test-time scaling is usually treated as a one-dimensional problem: decide how much more reasoning, sampling, search or verification to perform. State construction creates a second inference action. We formalize the resulting two-axis allocation problem under one budget and report a preregistered controlled run over 16 held-out generated families. The frozen joint arm scores 0.8582, versus 0.4631 and 0.4528 for the two named one-axis arms, and the run is byte-reproducible. A later comparator-capability audit changes the interpretation, not the historical bytes: the joint arm can emit four allocations while each one-axis arm can emit only two, and both losing-arm ceilings are below the winner's achieved score. With identical four-action sets, the gain is +0.0408, the worst-family gain is +0.0020, and the original gate is not met. Consequently `P12A_COMPARISON_VALIDITY_ADJUDICATION_V1.json` sets the active terminal to `P12A_SUPERIORITY_AUTHORITY_WITHHELD`. P12A motivates the resource-location hypothesis but does not establish signal-count superiority; that claim requires a prospectively frozen, capability-matched P12B.

1 Introduction

Test-time computation has become an explicit design variable in modern reasoning systems. Systems allocate more tokens, samples, search nodes, verifier calls or iterative refinement to difficult instances. Recent work develops bandit, constrained-policy and learned adaptive allocation strategies, reinforcing a general lesson: uniform inference budgets waste computation when item difficulty is heterogeneous.

But difficulty itself has more than one source. Some tasks are difficult because the relevant structure is poorly exposed in the current representation. Others are difficult because substantial search or reasoning remains even after the right structure is visible. If a system spends all marginal budget on reasoning in an access-limited task, it reasons harder over the wrong state. If it spends all marginal budget on state construction in a reasoning-limited task, it repeatedly reorganizes information that was already accessible.

P12 asks:

> **Under one matched total budget, when should a system spend computation changing state, when should it spend computation reasoning over state, and can a prospective policy learn or exploit the difference?**

The system is modeled as

`raw/current state -> optional state construction -> downstream reasoning/search -> verified outcome`.

The paper makes three contributions.

1. **Two-axis inference formulation.** State construction and downstream reasoning are symmetric budgeted actions rather than free preprocessing plus paid reasoning.
2. **A strict comparator contract.** The joint policy must beat both adaptive-state-only and adaptive-reasoning-only policies at identical total resources, not merely a fixed baseline.
3. **A comparator-capability correction.** The protected run is reported with a later hostile adjudication showing that signal count and permitted allocations varied together; a capability-matched P12B is therefore required.

The historical result is intentionally controlled and exactly reproducible, but its superiority interpretation is withheld by `P12A_COMPARISON_VALIDITY_ADJUDICATION_V1.json`. Active terminal: `P12A_SUPERIORITY_AUTHORITY_WITHHELD`. Its current value is to expose the stronger contract a successor must satisfy: equal total budget, equal action capability, and only then a variation in available signals.

2 Donor boundary and novelty

2.1 Adaptive test-time compute is prior-owned

Recent systems allocate inference compute dynamically based on predicted difficulty, value or resource constraints. Bandit formulations, constrained policy optimization, adaptive demonstration/generation strategies and "when to think" policies already own the primitive that different examples deserve different reasoning budgets. P12 therefore does not claim adaptive inference allocation itself.

2.2 Dynamic state construction is also prior-owned

Retrieval, compression, context selection, query-conditioned memory and structured-state construction already adapt what a model sees. P11 additionally supplies controlled evidence that construction can change accessibility. P12 does not claim dynamic state selection as a new primitive.

2.3 Residual after subtraction

The live residual is the **competition between those actions under one resource boundary**:

> State construction and downstream reasoning are two places to spend test-time computation. A valid joint-allocation result must hold total resource fixed and strictly improve over policies allowed to adapt either axis alone.

In the current donor set, adaptive-compute methods optimize downstream reasoning/sampling/search or generation control; they do not make costed state construction and downstream reasoning symmetric decision variables under the same matched envelope and then require superiority over both one-axis adaptive controls.

3 Formal problem and why joint allocation can be strictly valuable

For item `i`, let:

- `R_i` be current/raw state;

- `c_i` be resource spent constructing/restructuring state;
- `r_i` be downstream reasoning/search resource;
- `B_i` be the total envelope;
- `z_i` be information available before the protected outcome;
- `Y_i(c_i,r_i)` be verified success or quality.

The policy chooses `(c_i,r_i)` using `z_i`, subject to the common accounting contract. In the controlled benchmark the budget is scalar and exact: `c_i+r_i<=B`. In a real system the resource is a vector and comparison is Pareto-based unless a cost scalarization is frozen before protected outcomes.

3.1 Policy classes

- `FIXED_STATE_FIXED_COMPUTE`: fixed allocation across items.
- `ADAPTIVE_STATE_ONLY`: may change state budget but not reasoning budget.
- `ADAPTIVE_REASON_ONLY`: may change reasoning budget but not state budget.
- `JOINT_STATE_REASONING`: may choose both under the same total envelope.
- `ORACLE_JOINT`: hindsight ceiling used only diagnostically.

Define
`joint_gain(B) = Q_joint(B) - max(Q_state_only(B), Q_reason_only(B))`.
A positive P12 result requires `joint_gain>0` under the frozen comparison, not merely superiority to a fixed policy.

Consider a one-unit world containing two prospectively distinguishable regimes. In the access-limited regime success requires spending the unit on state construction; in the reasoning-limited regime success requires spending it on reasoning. A state-only adaptive policy cannot solve the latter and a reasoning-only adaptive policy cannot solve the former. A joint policy that sees the regime signal can spend the same unit at the valuable locus.

This existence argument is elementary and is not the empirical contribution. Its purpose is to identify the condition P12 must test: **heterogeneity in the location of marginal computation value**.

4 Protected matched-budget benchmark

4.1 Resource regimes

Each protected item has one hidden requirement:

- `EASY = (0,0)`;
- `ACCESS = (2,0)`;
- `REASON = (0,2)`;
- `BOTH = (1,1)`.

Every policy receives total budget `B=2`. Success is exact: allocated state and reasoning resources must meet both requirements. No arm receives extra budget and unused resource is not retrospectively reassigned.

4.2 Held-out families

The protected split contains **16 held-out families × 512 items**. Family regime proportions vary, while a uniform mixture component prevents degenerate single-regime families. Signal noise `sigma_f` ranges from `0.30` to `0.80` across families.

All adaptive arms receive the same pre-outcome signals

`s_c = c_req + Normal(0,sigma_f)s_r = r_req + Normal(0,sigma_f).`

These signals contain no protected success outcome, verifier result or post-allocation feedback.

4.3 Frozen policies

- `FIXED_11`: always `(1,1)`.
- `ADAPTIVE_STATE_ONLY`: choose `(2,0)` if `s_c>=1`, else `(0,0)`.
- `ADAPTIVE_REASON_ONLY`: choose `(0,2)` if `s_r>=1`, else `(0,0)`.
- `JOINT_FROZEN`: choose the feasible allocation in {`(0,0)`,`(1,1)`,`(2,0)`,`(0,2)`} nearest to `(s_c,s_r)` under squared Euclidean distance, with frozen tie order.
- `ORACLE_JOINT`: exact hindsight requirement, diagnostic only.

No policy is tuned on protected family outcomes.

5 Results

The historical protected terminal is `P12A_JOINT_ALLOCATION_SUPERIORITY_SUPPORTED`. Current claim authority is `P12A_SUPERIORITY_AUTHORITY_WITHHELD` under `P12A_COMPARISON_VALIDITY_ADJUDICATION_V1`.

policy	mean verified success
`JOINT_FROZEN`	**0.858154**
`FIXED_11`	0.515503
`ADAPTIVE_STATE_ONLY`	0.463135
`ADAPTIVE_REASON_ONLY`	0.452759

The joint policy improves over the better one-axis adaptive policy by **mean** `+0.334717`, family-block 95% bootstrap CI `[0.286008, 0.382693]`.

The **worst held-out family gain is** `+0.158203`. Joint versus fixed `(1,1)` gain is `+0.342651` on average. Every allocation respects the two-unit budget, the oracle ceiling holds in every family, and two fresh executions produce the identical SHA-256

`0194bc094f5696583533af5baae41e7c339902603d3706c8a1d2a78493f98947`.

5.1 Why the historical contrast is not a signal-count result

Both one-axis policies use pre-outcome signals, but they may emit only two allocations while `JOINT_FROZEN` may emit four. `ADAPTIVE_STATE_ONLY` has a perfect-signal ceiling of 0.475464 and `ADAPTIVE_REASON_ONLY` a ceiling of 0.463623, both below the winner's achieved 0.858154. The comparison therefore does not isolate the value of a second signal.

5.2 Regime interpretation

The large gain arises primarily because a one-axis policy's restricted action set cannot serve the opposite-axis and jointly limited regimes at any signal value.

Giving one-signal policies the same four allocations yields mean gain 0.040771, family-block interval [0.031006, 0.050659], and worst-family gain 0.001953. The original positive gate then returns `P12A_JOINT_ALLOCATION_SUPERIORITY_GATE_NOT_MET`. A new frozen P12B, not a post-hoc threshold change, is required for positive authority.

6 Resource-accounting contract and statistical analysis

The scalar controlled budget is intentionally clean. Real systems require vector receipts including:

- compiler/retrieval/preprocessing model and operations;
- state tokens/bytes and memory traffic;
- downstream generated tokens/recurrent steps;
- search nodes and verifier calls;
- tool calls and external latency;
- cache/recovery cost;
- model identity/capacity;
- end-to-end latency and reproducible energy where available.

A joint policy cannot "win" by shortening the downstream trace while hiding expensive retrieval or compilation upstream. Likewise a reasoning-only baseline cannot receive a larger model or search cap. If resource vectors are incomparable, the result should be a quality–resource Pareto frontier rather than a post-hoc weighted score.

6.1 Statistical analysis

The protected unit of generalization is the held-out family. Policies see paired items inside each family, while headline uncertainty is computed by a deterministic **20,000-resample family-block bootstrap** over the 16 family-level joint gains. The registered lower bound is positive.

The analysis does not pool items as if 8,192 individual trials were independent domains. Hyperparameters are frozen before protected evaluation. The worst-family gain is reported to prevent a favorable mean from hiding a family-level failure.

7 Relation to current adaptive-compute literature and limitations

Strategic test-time-compute allocation treats inference budget as a learnable or bandit decision across examples. Constrained policy approaches optimize accuracy under average compute. Adaptive in-context demonstration and generation methods jointly alter conditioning and generation effort. Recent "when to think" work likewise emphasizes selective reasoning to reduce unnecessary inference.

These results sharpen P12's motivation: **adaptive inference is crowded; the novel discriminator must be where the resource can be spent.** P12A does not yet measure that discriminator because the action portfolio changes with signal count. `P12A_COMPARISON_VALIDITY_ADJUDICATION_V1.json` therefore records `P12A_SUPERIORITY_AUTHORITY_WITHHELD`.

7.1 Limitations and real-system promotion gate

1. The protected benchmark is a controlled resource world, not an LLM, prover or production agent.

2. The pre-outcome signals are constructed measurements of resource need. Real signal quality may be substantially worse.
3. Scalar units are commensurate by construction. Real compiler work, tokens, verifier calls and latency are heterogeneous.
4. The joint policy is a simple frozen nearest-allocation rule; the paper does not claim it is optimal.
5. P12B must first give one- and two-signal policies an identical allocation set; a real-system result must additionally include strong compute-only and state-only adaptive baselines.
6. A broad superiority claim requires at least one held-out real LLM/procedural domain or verifier-backed search domain under matched end-to-end resource receipts.
7. If real tasks overwhelmingly favor one resource locus, a simpler one-axis policy may be preferable; P12 predicts this as a regime condition rather than denying it.

8 Discussion and conclusion

P12 reframes test-time scaling as a **portfolio of computations**. "Think longer" is not the only adaptive action available to an intelligent system. It may be cheaper to parse, retrieve, compile, restructure or recover state so that less downstream search is required. Conversely, when state already exposes the relevant structure, additional preprocessing is wasteful and reasoning should receive the marginal budget.

The protected benchmark demonstrates the construction but does not establish the key causal discriminator. Equal total budget was real; equal action capability was not. Most of the margin was unreachable by the named baselines before their signals were read.

This leaves a concrete systems hypothesis for P12B and real agents: **test-time scaling curves may be two-dimensional, with state-work and reasoning-work measured on a common receipt and with action capability held fixed across signal ablations.**

Adaptive inference may need to decide not only **how much** computation to spend but **where** to spend it. P12 supplies the formulation and an exact failure analysis of its first empirical discriminator. Under `P12A_COMPARISON_VALIDITY_ADJUDICATION_V1.json`, the active terminal is `P12A_SUPERIORITY_AUTHORITY_WITHHELD`. The next step is a prospectively frozen P12B with identical four-action capability; real end-to-end validation follows only after that controlled contrast is sound.

9 References

- Zuo, B. & Zhu, Y. *Strategic Scaling of Test-Time Compute: A Bandit Learning Approach.* ICLR 2026.
- Zuo, B., Zhou, D. & Zhu, Y. *Adaptive Test-Time Compute Allocation with Evolving In-Context Demonstrations.* Findings of ACL 2026, 35156–35173. DOI: 10.18653/v1/2026.findings-acl.1754.
- Zhai, Z., Li, B., Xiao, B., Li, M. & Wang, X. *Adaptive Test-Time Compute Allocation for Reasoning LLMs via Constrained Policy Optimization.* arXiv:2604.14853, 2026.
- *Learning When to Think: Adaptive Reasoning for Test-Time Compute Allocation.* arXiv:2608.20256, 2026.
- P11 provides the controlled state-construction basis consumed by this paper; it does not transfer scientific authority to P12.